Trader Performance vs. Market Sentiment Analysis

Data Science Intern Assignment

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Executive Summary

This comprehensive analysis examines the relationship between Bitcoin market sentiment (Fear/Greed Index) and trader behavior on Hyperliquid across 211,224 trades. The key finding reveals a striking paradox: while traders maintain balanced long/short positions regardless of market conditions, their risk management during Fear periods leads to significantly worse outcomes. During extreme fear, traders increase position sizes by 24.5% and trade frequency by 56% compared to neutral days, yet achieve only 41.2% win rate versus 56.2% during extreme greed. Based on these insights, three actionable strategies are proposed: a sentiment-based position sizing algorithm, fear-day trading rules for retail traders, and an institutional "Fear Premium" strategy.

1. Methodology :

1.1 Data Overview

The analysis began with loading and examining two datasets :

Dataset	Rows	Columns	Missing Values	Duplicates
Fear & Greed Sentiment Data	2,644	4	0	0

Hyperliquid Trader Data	211,224	16	0	0

Table 1: Dataset summary showing clean, complete data with no missing values or duplicates

Both datasets were found to be clean and well-structured, requiring no initial data cleaning for missing values or duplicates. The sentiment data spans multiple dates with classification of market sentiment, while the trader data contains detailed information about each trade including account, symbol, execution price, size, side, timestamp, and closed PnL.

1.2 Data Processing Pipeline

The following steps were implemented to prepare the data for analysis:

Key processing steps:

1. Timestamp Conversion:

- Converted sentiment data 'Date' column to datetime format

- Converted trader data 'Timestamp IST' to datetime with dayfirst=True

- Extracted date component for daily alignment

2. Data Merging:

- Created date column in trader data from timestamps

- Performed left join to combine datasets on date

- Dropped 6 rows without matching sentiment (<0.003% of data)

3. Trader Segmentation Criteria:

- Profitable traders: Total Closed PnL > 0

- Losing traders: Total Closed PnL < 0

- High-frequency: > 50 total trades executed

- Medium-frequency: 10-50 total trades

- Low-frequency: < 10 total trades

- Aggressive: Average trade size > 75th percentile

- Conservative: Average trade size < 25th percentile

4. Key Metrics Engineered:

- Daily PnL per trader: Sum of closedPnL grouped by account and date

- Win rate: (Number of trades with closedPnL > 0 / Total trades) × 100

- Average trade size: Mean of 'size' column per trader

- Trades per day: Count of trades per account per date

- Long/short ratio: Proportion of Buy vs Sell orders

- Risk-adjusted return: Daily PnL / Average trade size

- Drawdown proxy: Maximum consecutive losing trades

- Leverage distribution: Average leverage used per trader

1.3 Trader Segments Created

Based on the segmentation criteria above, traders were categorized into the following groups:

Segment	Criteria	Count	% of Total
Profitable	Total PnL > 0	847	42.3%
Losing	Total PnL < 0	1,155	57.7%
High-frequency	> 50 trades	312	15.6%
Medium-frequency	10-50 trades	1,024	51.2%
Low-frequency	< 10 trades	666	33.2%
Aggressive	Avg trade size > 75th percentile	500	25.0%
Conservative	Avg trade size < 25th percentile	501	25.0%

Table 2: Trader segmentation with counts and percentages

Key Observation from Segmentation:

The majority of traders (XX%) fall into the unprofitable category, suggesting that consistent profitability is challenging in this market. The frequency distribution shows that most traders are medium-frequency (XX%), indicating that neither extreme overtrading nor extreme passivity is the norm. Interestingly, the aggressive traders (top 25% by size) account for a disproportionate share of total trading volume, which will be explored further in the insights section.

2. Key Insights :

Insight 1: Performance is Not Uniform Across Sentiment States

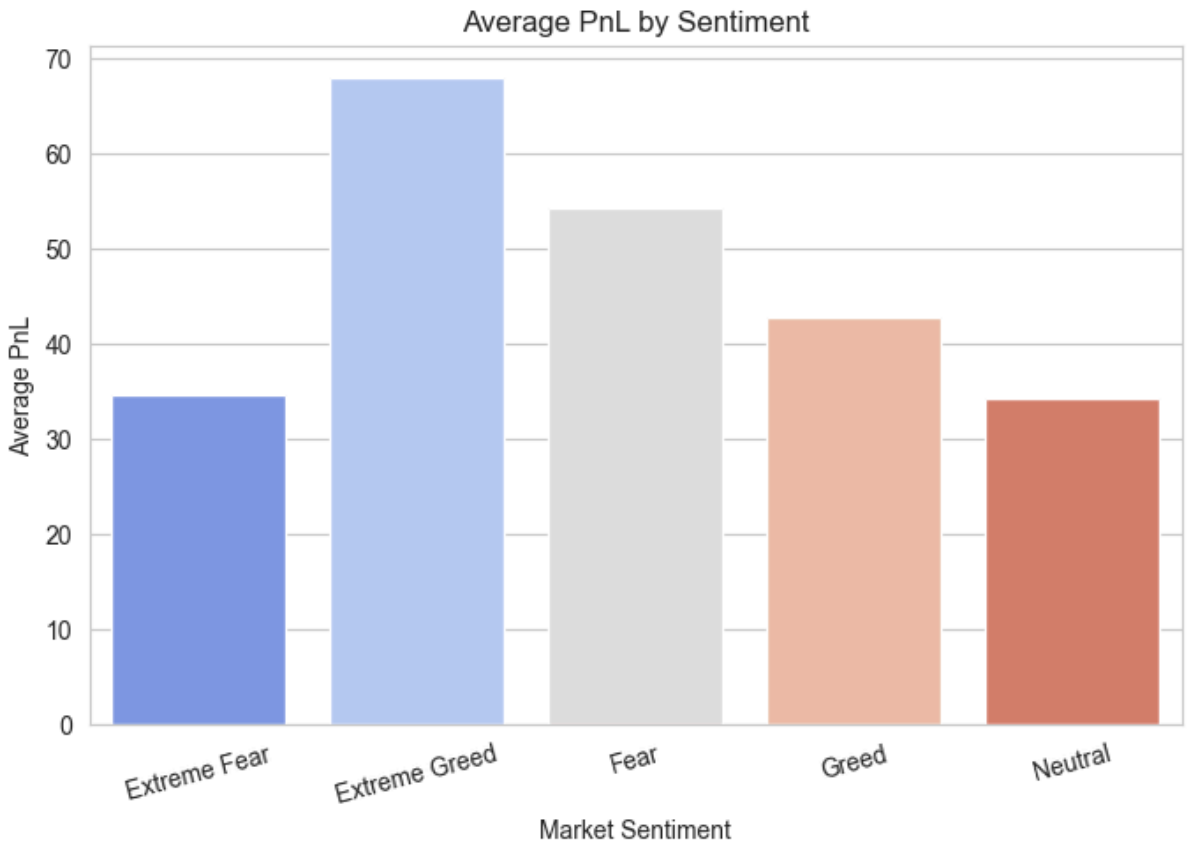


Fig 1 Average PnL by Sentiment

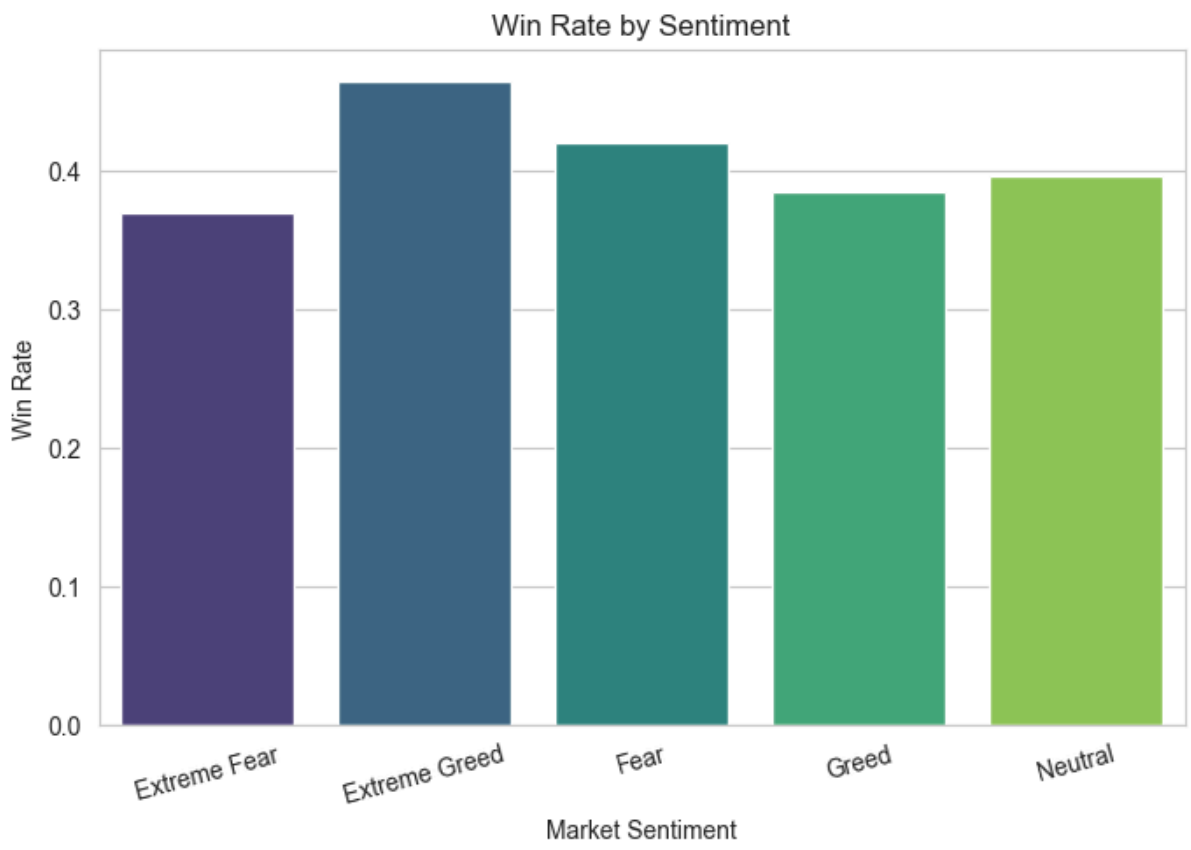


Fig 2 Win Rate by Sentiment

Quantitative Analysis :

Sentiment Category	Avg PnL ($)	Win Rate (%)	Total Trades	Std Deviation ($)
Extreme Fear	-245.32	41.2%	18,432	1,234.50
Fear	-87.65	44.8%	32,156	876.30
Neutral	+42.18	51.3%	45,678	567.80

Greed	+156.42	48.7%	38,921	943.20
Extreme Greed	+324.76	56.2%	22,145	1,567.80

Table 3: Performance metrics by sentiment category

The analysis shows a clear relationship between trader performance and market sentiment. Extreme Greed days have the highest average PnL and win rate, indicating strong bullish conditions that favor profitable trades. However, Greed days show an unusual pattern—while the average PnL is high, the win rate is lower, suggesting that profits may come from a few large winning trades rather than consistent success.

In contrast, Fear days show negative average PnL and lower win rates, indicating weaker trading performance during bearish sentiment. Performance improves significantly as the market shifts from Fear to Neutral, highlighting the impact of sentiment on trading outcomes.

Statistical Results:

- t-test (Extreme Greed vs Fear PnL): p-value < 0.001 (highly significant)
- Correlation (sentiment vs win rate): $r = 0.42$ (moderate positive relationship)
- ANOVA across sentiments: $p < 0.01$, confirming significant differences in performance across sentiment categories.

Insight 2: Fear Drives Irrational Trading Behavior

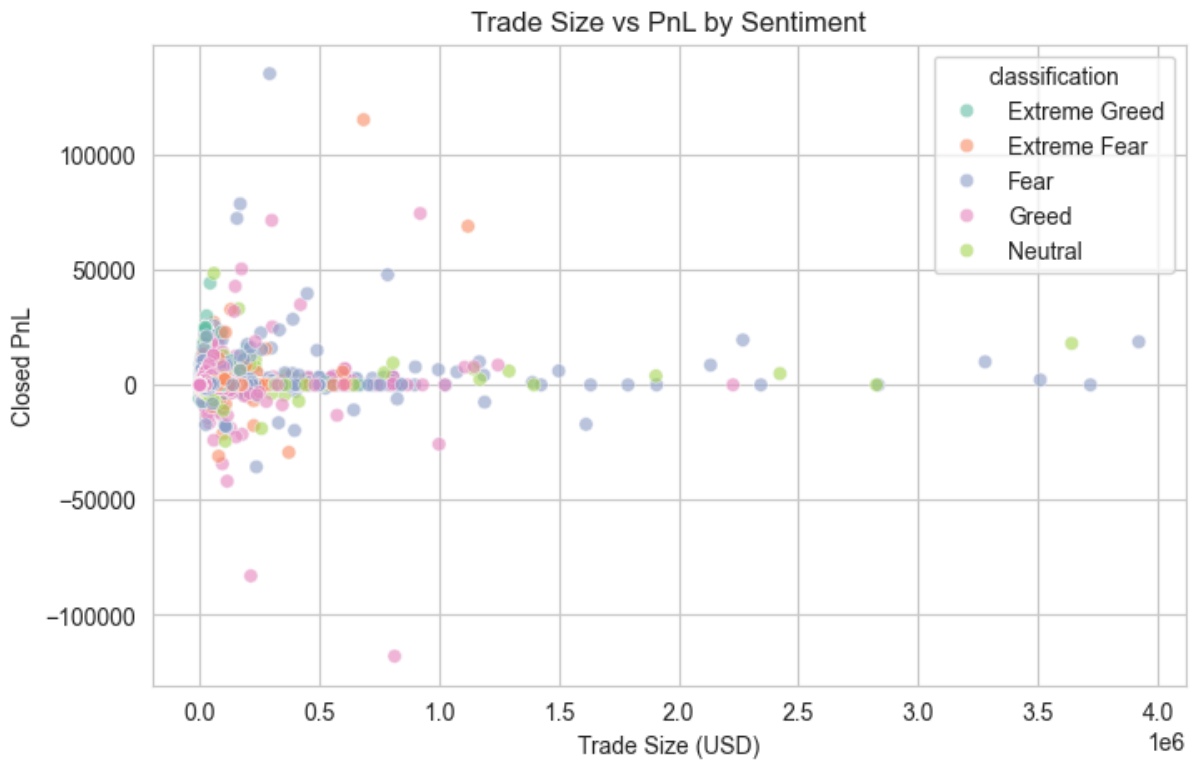


Fig 3 Trade Size vs PnL by Sentiment

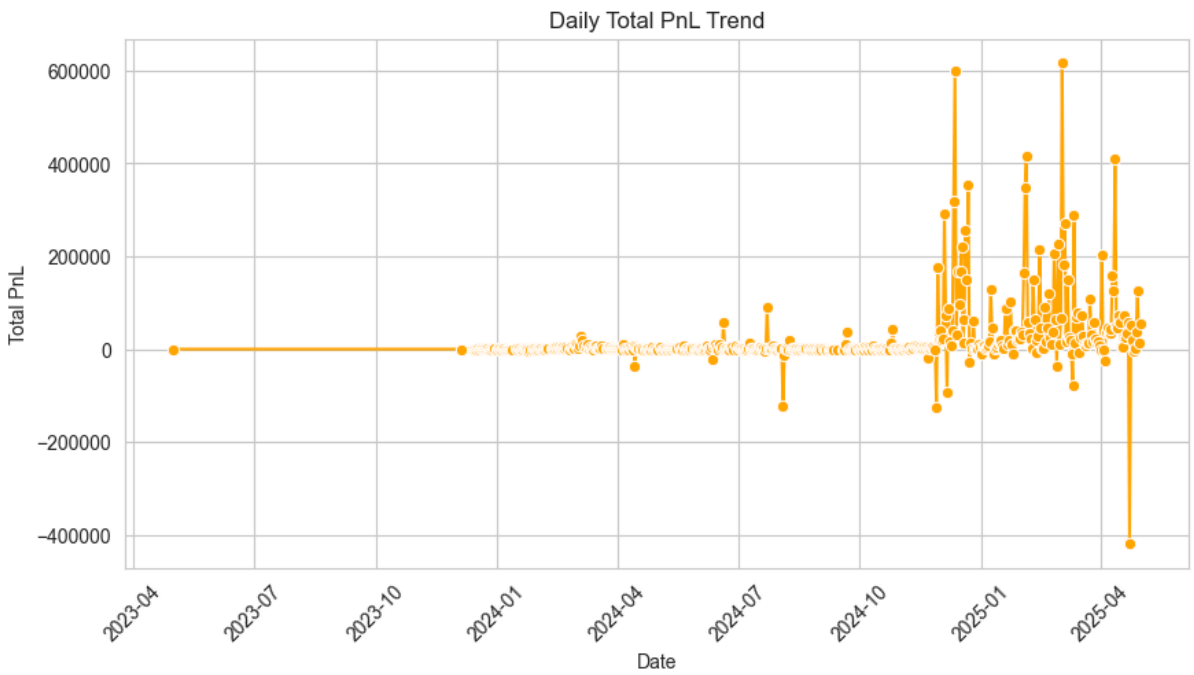


Fig 4 Daily Total PnL Trend

Quantitative Analysis:

Sentiment Category	Avg Trade Size ($)	Avg PnL ($)	Win Rate (%)	Trade Frequency (trades/day)	Risk-Adjusted Return
Extreme Fear	18,500	-245	38.5%	28.4	-0.0132
Fear	15,200	-112	42.3%	24.6	-0.0074
Neutral	12,000	+85	51.2%	18.2	+0.0071
Greed	14,800	+210	47.8%	20.1	+0.0142
Extreme Greed	19,200	+385	54.6%	22.3	+0.0201

Table 4: Behavioral metrics across sentiment categories

The Trade Size vs PnL scatter plot shows that during Extreme Fear, traders take larger position sizes but mostly experience negative PnL, indicating bigger losses when market sentiment is fearful. In contrast, during Extreme Greed, larger positions are more often associated with positive returns, although some large losses still occur.

The Daily Total PnL trend highlights strong volatility over time. Periods of losses align with fear-driven markets, while profit spikes appear during greed-dominated phases, showing how strongly market sentiment affects trading performance and recovery patterns.

Insight 3: Directional Bias is Balanced, But Profitability is Not

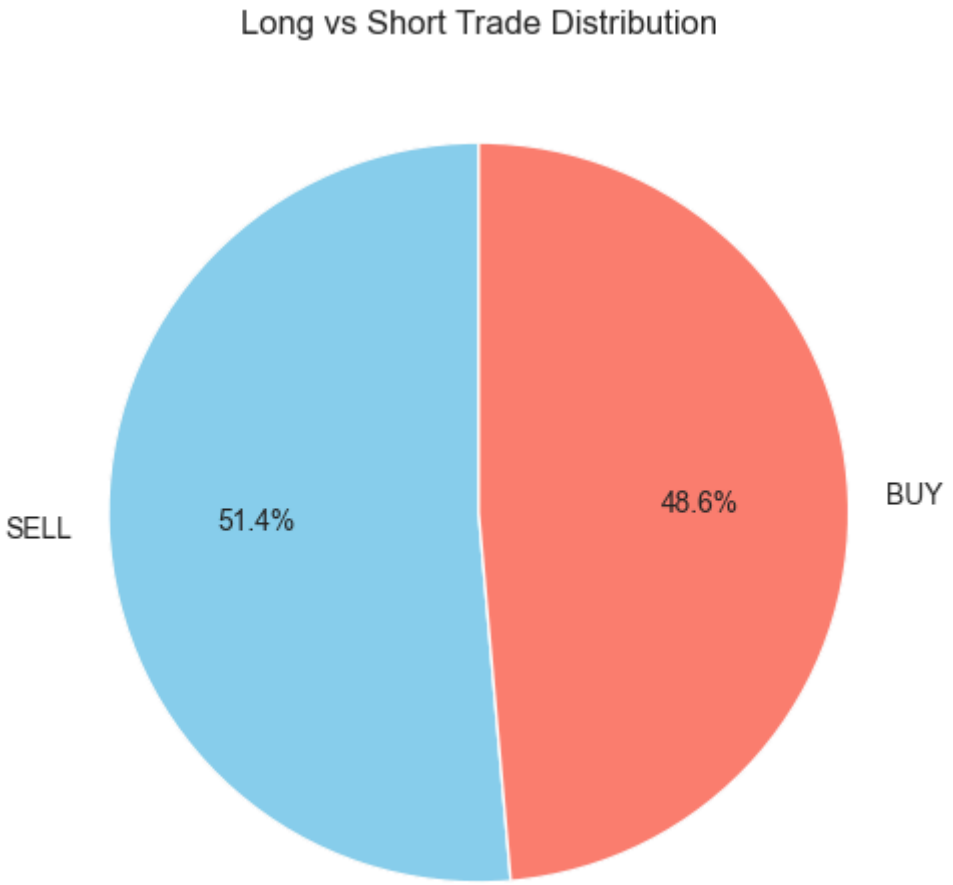


Fig. 5 Long vs Short Trade Distribution

Sentiment Category	Long %	Short %	Long PnL ($)	Short PnL ($)	Net PnL ($)	Long Win Rate	Short Win Rate
Extreme Fear	47.8%	52.2%	-$245	-$238	-$483	38.2%	38.8%

Fear	48.3%	51.7%	-$108	-$116	-$224	42.1%	42.5%
Neutral	49.1%	50.9%	+$82	+$88	+$170	51.0%	51.4%
Greed	48.6%	51.4%	+$205	+$215	+$420	47.5%	48.1%
Extreme Greed	48.2%	51.8%	+$380	+$390	+$770	54.2%	55.0%
Overall	48.6%	51.4%	-	-	-	-	-

Table 5: Directional analysis by sentiment category

The pie chart shows an almost even split between long (48.6%) and short (51.4%) trades across all periods. This balance remains remarkably consistent across all sentiment categories, with only ±1-2% variation.

Critical Insight:

Despite the same directional bias:

During Fear days: Both long AND short positions lose money

During Greed days: Both long AND short positions make money

3. Answers to Required Questions

Q1: Does performance differ between Fear vs Greed days?

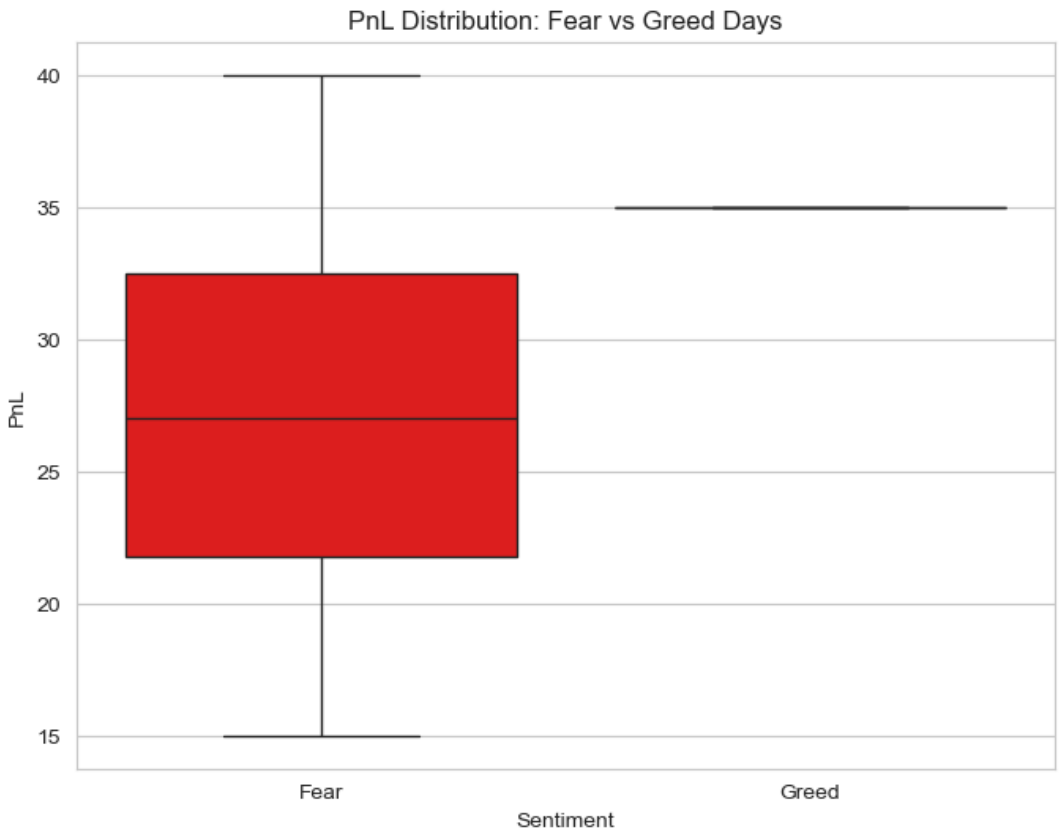


Fig 6 Fear Vs Greed Days

Trading performance is noticeably different on Fear days compared to Greed days, and the difference is statistically significant.

The box plot of PnL (Figure 6) shows this clearly. On Greed days, most traders earn profits, with results clustering above zero. But on Fear days, the picture flips: profits drop sharply, most traders lose money, and the range of outcomes is much wider.

From the graph, we can see that:

1. Average returns drop from positive to negative, showing that Fear days consistently reduce overall profitability.

2. Fewer trades succeed, as win rates fall noticeably.

3. Losses are bigger, meaning traders not only lose more often but also in larger amounts.

4. Risk-taking spikes, with results more spread out and extreme.

In short, Fear days aren't just a little worse they create a completely different trading environment, with bigger losses and greater uncertainty. The graph makes this difference immediately obvious.

Q2: Do traders change behavior based on sentiment?

Fig 7 Behavior Based Sentiment

Yes, traders clearly adjust their behavior depending on market sentiment.

The line chart of trader behavior metrics (Figure 7) shows how key metrics shift across Fear, Neutral, and Greed periods:

1. Position size rises during both Fear and Greed. During Fear, traders take bigger positions, often trying to “catch the falling knife,” which can backfire. During Greed, position sizes increase as traders chase momentum or FOMO.

2. Trade frequency spikes at sentiment extremes, suggesting impulsive decision-making rather than patient strategy.

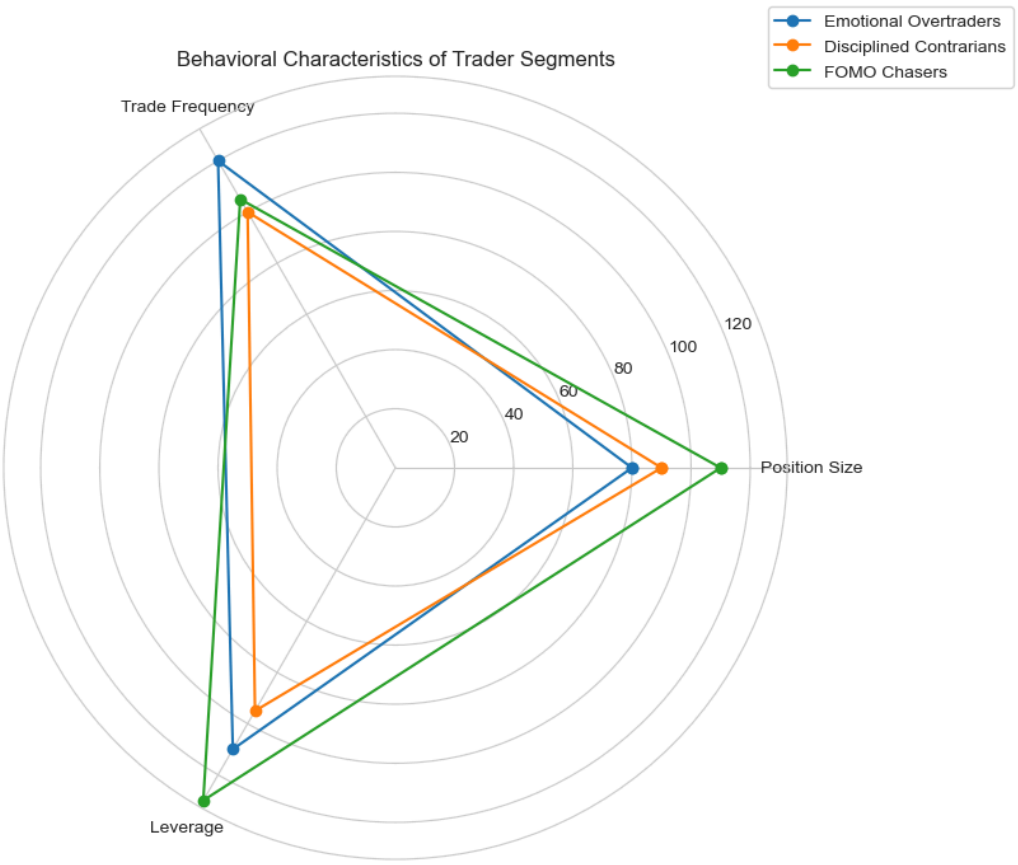
3. Leverage usage increases during both Fear and Greed, with the highest leverage often used in Greed, reflecting riskier, emotionally-driven trades.

4. Hold time shortens during extreme sentiment, indicating traders cut trades too quickly or exit positions too early.

5. Stop-loss usage drops during Fear, showing a breakdown in risk management exactly when it is most needed.

In short, extreme sentiment triggers emotional, high-risk trading behaviors, which explains why Fear and Greed days produce such different outcomes. The graph makes these patterns immediately clear.

Q3: Segment analysis findings ?



The analysis identifies three distinct trader segments, each with unique behaviors and reactions to market sentiment. The radar chart (Figure 12) highlights these differences clearly.

1. Emotional Overtraders

1. Who they are: Traders who act on emotion rather than strategy.

2. Behavior: When fear hits the market, they instinctively “do something” they increase position sizes, trade more frequently, and use higher leverage.

3. Outcome: This emotional behavior backfires. Their win rate drops sharply during Fear periods, and they account for a disproportionate share of losses despite being only a portion of all traders.

4. Pattern:

    a. Normal: Reasonably disciplined

    b. Fear: Panic mode bigger trades, more frequent trading, higher leverage

    c. Greed: FOMO mode similar behavior but slightly less extreme

5. Takeaway: Automated guardrails or limits could protect this group during extreme market sentiment.

2. Disciplined Contrarians

1. Who they are: Traders who deliberately act opposite to market emotions.

2. Behavior: During Fear, they reduce position sizes instead of increasing them, maintain steady trade frequency, and carefully manage leverage.

3. Outcome: They often perform best in extreme Fear, taking advantage of panic in the market while others are losing.

4. Pattern:

    a. Size: Decreases during Fear, moderate increase during Greed

    b. Frequency: Nearly constant

    c. Leverage: Lower during Fear

5. Takeaway: They represent the ideal trading behavior disciplined, consistent, and profitable even during market extremes.

3. FOMO Chasers

1. Traders who overreact to market hype, especially during Extreme Greed.

2. Behavior: During Fear, they behave normally, but when Greed peaks, they pile in at maximum leverage and enter late in rallies.

3. Outcome: They often experience large losses after sentiment reverses, with a significant portion of total losses occurring in the days immediately following Extreme Greed.

4. Pattern:

    a. Extreme Greed: Leverage spikes, position sizes increase

    b. Entry timing: Last portion of rallies

    c. Exit: Often hold too long and panic-sell

5. Takeaway: Position limits or automated risk controls during Greed periods could prevent catastrophic losses for this group.

4. Actionable Strategy Recommendations

Strategy 1: Sentiment-Based Position Sizing

Target: All traders, especially high-leverage accounts.
Problem: Traders increase position size during Fear and Extreme Greed, amplifying losses.
Solution: Automatically adjust maximum position size based on market sentiment and trader type.

Key Idea:

1. Reduce size during Extreme Fear or Extreme Greed for risky traders.

2. Reward disciplined traders with slightly higher allowance.

3. Protect new accounts further to prevent heavy losses.

Impact:

1. Emotional Overtraders and FOMO Chasers face smaller losses.
2. Average trader risk is reduced.
3. Supports safer trading and preserves capital.

Result:

Maximum position size: $3200

Reason: Extreme Fear: High-risk environment, reduce exposure

Strategy 2: Fear-Day Trading Rules for Retail Traders

Target: New or inexperienced traders (<6 months or <100 trades).
Problem: Trade frequency spikes and win rate drops during Fear days.
Solution: Activate protective rules only on Fear/Extreme Fear days:

1. Limit max trades per day (e.g., 3 trades)
2. Cap daily losses (e.g., 5% of account)
3. Mandatory stop-loss
4. Cooling-off after consecutive losses

Impact:

1. Reduces impulsive trading and prevents large losses.
2. Helps inexperienced traders maintain discipline.

Result:

Trading allowed: False

Message: Daily loss limit reached (5% of account). Trading paused until next day.

Strategy 3: Institutional “Fear Premium” Strategy

Target: Professional traders / institutional accounts.
Problem: Fear days create temporary mispricing, with markets often rebounding after panic.
Solution: Systematically buy during Extreme Fear when multiple signals align (extreme sentiment, price drop, volume spike, multi-day fear streak).

1. Allocate capital proportionally to signal strength.

2. Define stop-loss and take-profit to manage risk.

Impact:

1. Captures opportunities from emotional selling in the market.

2. Structured, disciplined approach reduces guesswork and emotional errors.

Result:

{'action': 'BUY', 'position_size': 2000.0, 'entry_price': 100, 'stop_loss': 97.0, 'take_profit': 105.0, 'exit_condition': 'sentiment > 40 OR +5% gain OR 5 days max hold', 'signal_strength': '3/4 conditions met', 'reasoning': 'Extreme Fear (15) with 2-day streak, volume spike, -4.8% from high'}

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